

A Grammar-Driven Rule-Based Stemmer for Pashto

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ABSTRACT

Pashto has roughly forty million speakers and very few text-processing tools. The only stemmer published for the language applies nine affix rules without a lexicon and without any means of choosing between competing analyses. We present a rule-based Pashto stemmer built from an affix inventory of 118 rules drawn from published grammars, in which every rule records its source, its productivity, and whether it may be applied at all. Candidate stems are generated separately from the decision of which candidate to keep, so a shallower analysis can displace a deeper one. No machine learning is used.

Since no annotated set for Pashto stemming existed, we built two. A development set of 2,912 word types and a held-out set of 500 types sampled from news text were each annotated independently by three native-speaker Pashto educators, who agreed on 96.88% of stems ($\kappa = 0.969$). The held-out set was annotated with its reference column empty and was not consulted while the system was being developed. 499 of its 500 types fall within the stemming task and are the denominator below; the remaining one needs a different word and is reported apart.

On that set the proposed system reaches 72.75% exact-match accuracy against 40.48% for a faithful reimplementation of the prior system, and 72.8% against 16.9% on the types that carry an affix. Paice's under-stemming index falls from 0.889 to 0.500.

A secondary finding concerns evaluation itself. The protocol used in the prior work — presenting system output to native speakers and asking whether it is acceptable — assigns a score of 100% to a system that modifies no word at all. We report results under both that protocol and exact match against a reference fixed in advance, and recommend the latter for Pashto stemming.

Keywords: Pashto, stemming, morphological analysis, rule-based methods, low-resource languages, evaluation methodology, information retrieval

1. INTRODUCTION

A reader searching for documents about houses does not distinguish between house, houses and of the houses. A retrieval system does, because to the system these are three unrelated strings. Stemming removes that difference by stripping the endings a word carries, so that related forms collapse onto a single index term. For English the problem has been settled since Porter's algorithm [2], which remains in production use after four decades.

For Pashto almost nothing exists. The language is spoken by approximately forty million people across Afghanistan and Pakistan, is one of the two official languages of Afghanistan, and has a growing volume of digital text in news and social media. It also has a rich inflectional system, which is precisely the condition under which stemming matters. A Pashto retrieval system without stemming will miss most relevant documents; a classifier will treat the singular and plural of a key term as

unrelated features and learn each from less data than it has.

The single published Pashto stemmer, by Aslamzai and Saad [1], applies nine affix rules in a fixed order with no lexicon and no selection stage. Our reimplementation of those rules leaves most words untouched, and damages a substantial fraction of the words it does modify.

Three properties of Pashto make a naive approach fail.

A. Five letters for one sound

Pashto writes five distinct characters conventionally called *ye*. They are not interchangeable; they carry gender, number and case:

TABLE I. THE FIVE PASHTO YE LETTERS

Letter	Example	Function
ی	سری	masculine singular direct
ي	سري	masculine plural/oblique; 3rd person present
ې	ښکلې	feminine; plural
ۍ	نجلۍ	feminine singular, one noun class
ئ	ووايئ	2nd person plural imperative

Normalization pipelines written for Persian or Urdu routinely merge these into a single character. For Pashto that destroys the information a stemmer needs: *سری* (man) and *سري* (men) become indistinguishable. Our system preserves all five.

B. Endings that are not endings

The same final letter may be an affix in one word and part of the root in another, with nothing in the surface form to separate the two cases: *لنډه* reduces to *لنډ*, where *ه* marks agreement, while *سيمه* must remain *سيمه*, because there the *ه* belongs to the word. A native speaker resolves this by knowing the word; a rule conditioned on the surface form cannot.

C. Verbs whose stems cannot be derived

Pashto distinguishes weak verbs, whose stems follow from the infinitive, from strong verbs, whose present stem must be known: *ليدل* has the present stem *وين*, and *تلل* has *خ*. No affix rule derives *وين* from *ليدل*. These verbs are frequent and must be listed.

D. Contributions

- An affix inventory of 118 rules in which each entry records its grammatical source, its productivity, a confidence level, and an explicit decision about whether it may be stripped. Twenty-seven entries are documented but never applied.
- Two annotated evaluation sets for Pashto stemming, one of which was annotated blind and never used during development.
- An evaluation protocol that separates performance on word types carrying an affix from performance on types that do not, because overall accuracy alone rewards inactivity in a language where nearly half of word types are uninflected.
- A quantitative account of what each component contributes, including two components that we retain

despite a measured cost and one rule that we implemented, measured and removed.

- The observation that the evaluation protocol used in prior Pashto stemming work cannot distinguish a working stemmer from an inert one.

2. RELATED WORK

A. Rule-based stemming

Lovins [3] published the first English stemmer, a single-pass longest-match algorithm over 294 endings. Porter [2] replaced it with a five-step procedure whose conditions depend on a measure of syllable count. Its durability is instructive: the algorithm is small, fast, entirely inspectable, and makes no claim to produce real words. Porter maps both relational and relate to relat, which is not an English word. The output of a stemmer is a class label rather than a lemma, and confusing the two is a recurring error in work on morphologically rich languages.

Paice [4] contributed both an algorithm and, more consequentially for the present work, an evaluation method [5]. His under-stemming index counts related word pairs a system fails to conflate, and his over-stemming index counts unrelated pairs it wrongly merges. Because the two move in opposite directions, reporting both is more informative than accuracy alone.

B. Arabic-script languages

Khoja and Garside [6] built an Arabic root extractor that strips affixes and matches the remainder against trilateral patterns. Larkey, Ballesteros and Connell [7] showed that light stemming, removing only a short list of prefixes and suffixes, outperforms aggressive root extraction for retrieval. Their conclusion shaped the design adopted here: removing less, but removing it reliably, is preferable to removing more. For Persian, Sharifloo and Shamsfard [8] used a bottom-up parser over a morpheme lexicon. The pattern common to these systems is that a lexicon resolves ambiguity that rules alone cannot.

C. Pashto

Work on Pashto stemming is confined, to our knowledge, to Aslamzai and Saad [1]. Because that system is the baseline for everything reported here, we describe it in full.

Their algorithm applies nine rules, four removing suffixes and five removing prefixes, each with a length condition. Rule 1 removes one of eleven suffixes from words of five or more characters; Rule 2 replaces a final ي with ه in words of four or more; Rule 3 removes من; Rule 4 removes يز and appends ه; Rules 5, 6, 7 and 9 remove prefixes; and Rule 8 reduces a reduplicated word to one copy.

Reproducing the baseline exactly mattered, since a comparison against a misimplementation would be worthless. Our implementation reproduces all eight worked examples given in the paper. Four points in the publication required a decision, and we record them so that they can be checked:

- The order of prefix and suffix removal is not stated. We remove suffixes first, because the paper's own example *موتروان* → *موتر* only succeeds in that order; with prefixes first, the Rule 5 prefix م removes the first letter of *موتر*.
- Rule 3 is stated for words of “more than 5 characters” but its example *واکمن* has exactly five, so we implement the condition as five or more.
- The Rule 5 prefix list contains a bare م, which contradicts the *موتروان* example above. We exclude it, giving the baseline the reading its own example implies rather than the one that would lower its score.

- Rule 4 instructs the reader to remove يز and append ه. Its first example follows this, but its second gives *دوديز* → *دود* with no ه.

Rule 2 is also linguistically incorrect as published. It maps the masculine plural adjective *بنکله* to *بنکلي*, which is the feminine singular; Tegey and Robson [9] give the masculine singular *بنکلی* as the base form of this class. The rule is corrected in our inventory.

Finally, the reported accuracy of 87% is not comparable with the figures in this paper, and the difference lies in the protocol rather than in the algorithm. We return to this in Section VI.

D. Grammatical sources

The affix inventory was assembled from published descriptions rather than by inspecting a corpus. Table II lists the sources used and the tag each carries in the released inventory file.

Of the 118 rules, 112 cite a published source, located by chapter in Tegey and Robson [9] or by section in Penzl [12]. The remaining six — *ل, لو, نې, نى, نى, ستان* — are entries the engine never applies. For those no grammar is being appealed to: the affix is real Pashto morphology, the reason it is recorded is that a reader would otherwise wonder why it is absent, and the reason it is not applied is measured behaviour, given with each entry in Appendix A. Every rule the engine can actually fire cites a published source.

The provenance of each rule is recorded in Appendix A rather than summarised, so a reader can check any individual entry. One distinction a single citation count would hide is worth drawing: twenty-two of the cited rules are documented and never applied. For those the citation establishes that the affix exists, not that removing it is safe, and the decision not to strip rests on measured behaviour rather than on the grammar.

TABLE II. SOURCES FOR THE AFFIX INVENTORY

Source	Tag
Tegey & Robson, A Reference Grammar of Pashto [9]	[T&R]
Robson & Tegey, “Pashto” [10]	[R&T]
David, Descriptive Grammar of Pashto [11]	—
Naeem & Khan, derivational morphology [13]	[N&K]
Khan, Pashto diminutives [16]	[Khan23]
Penzl, A Grammar of Pashto [12]	[Penzl]
Wiktionary / kaikki.org Pashto suffixes	[W]

3. PASHTO MORPHOLOGY

This section covers the parts of the morphology that bear on the design. It is not a complete grammatical description.

A. Orthography

Beyond the Arabic inventory, Pashto adds retroflex and other consonants that occur in neither Arabic nor Persian: ت ډ ر ږ ځ ښ څ ښ ګ. A word containing any of these is certainly Pashto rather than a loan, and the system uses that fact.

A complication became apparent only late in this work. The same suffix is written with different ye letters in different registers: dictionary orthography prefers لومړی where news orthography writes لومړي. A rule spelled in one convention does not fire on the other. The system therefore compares a folded key when deciding whether a rule applies, while the letters it writes out are never altered. Section VII quantifies the effect.

B. Nominal inflection

Tegey and Robson divide Pashto nouns into masculine classes M1–M4 and feminine classes F1–F3, each with its own plural and oblique endings. The endings the system removes most often are ونه and ونو (masculine plural and its oblique), ې (animate plural and oblique), ان (feminine plural) and و (oblique plural). Nominal inflection is not a simple suffix-substitution system: gender, number, noun class and the final shape of the stem can all trigger stem alternation.

C. Verbal morphology

Two productive patterns form verbs from nouns and adjectives. The بدل intransitive and ول transitive patterns are regular, so both are handled by rule and neither appears in the dictionary: جورېدل → جور and جورول → جور. For strong verbs the present stem cannot be predicted, and these are listed: لیدل / ولین / تلل / کول / ځ / غواړل / غوښتل / کو.

D. Loanwords

Pashto has borrowed heavily from Arabic, Persian and English, and loanwords are the principal remaining source of error, because their endings frequently resemble Pashto affixes. The correct stem of منشي is منشي, not منش; of

کورس is کورس, not کور, which is a different word meaning *house*; and of بېجينگ (Beijing) is the word itself, not جينگ. Arabic broken plurals are a related problem, formed by changing the internal vowel pattern rather than by adding material, so that no suffix rule can reach them.

4. METHOD

Fig. 1 shows the order in which a word passes through the system. Three of the stages can return a stem on their own, and the remaining word goes to the generate-and-select stages that do the general work. Every stage can be disabled independently, which is what makes the component study in Section VII possible.

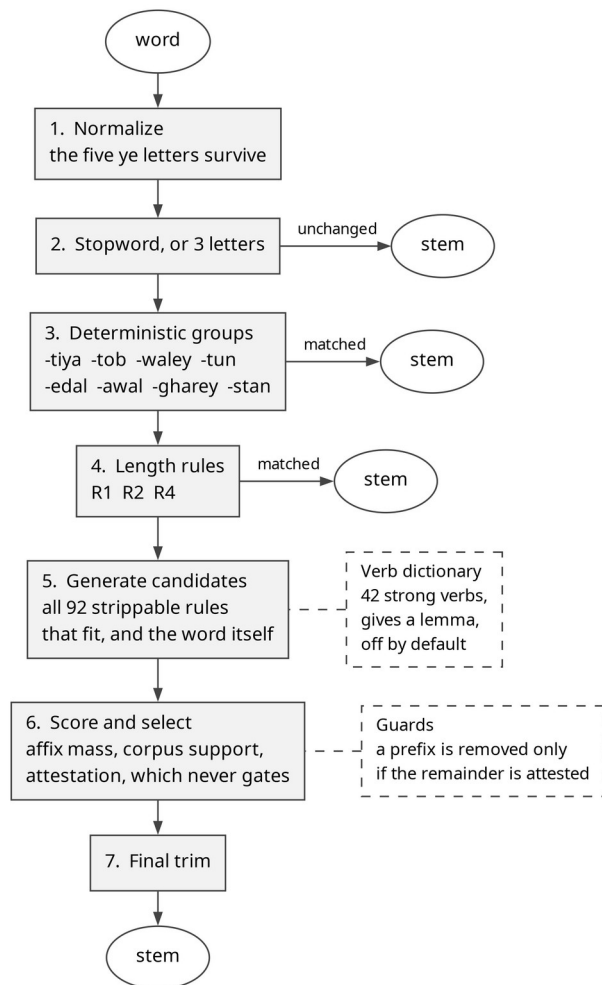


Fig. 1. The pipeline. Stages 2–4 can each return a stem and stop; everything else reaches stage 5, where all applicable rules are applied and the results are then scored against one another. The two dashed boxes are not stages: the guards constrain which strips the earlier stages may make, and the irregular-verb dictionary is an optional lemmatization mode that is off by default. Affixes are transliterated in the figure for legibility and given in Pashto script in the text.

A. Normalization

Normalization removes differences that carry no information: Arabic diacritics, the tatweel elongation character, zero-width joiners, and the several Unicode encodings of the same letter. The Arabic kaf is mapped to the Pashto keheh, the alef-hamza forms to bare alef, and the Urdu letters common in Pashto written in Pakistan (ٹ ٺ ڙ ځ and the Persian گ) to their Pashto counterparts. These are encoding corrections rather than changes of spelling.

What normalization does not do is merge the ye letters. We verified that no ye character is altered in any cell of either evaluation set. This holds through the whole pipeline, not only at the normalization step: the output is always a slice of the input token, so whichever ye the word was written with is the one the stem carries. A rule spelled with ی that fires on a word spelled with ې returns ې. The aggressive fold remains available behind a configuration flag so that the cost of the error reported in the literature can be measured rather than asserted.

B. The affix inventory

The inventory holds 107 suffix rules and 10 prefix rules. Each rule records the affix and the side it attaches to; whether it is inflectional or derivational; a minimum surviving stem length; the part of speech the affix signals, where known; its productivity (productive, restricted, lexicalized or borrowed); a confidence level; and whether it may be stripped at all.

Table III gives the schema in full, with three entries written out. The last field does real work. Twenty-seven entries are documented but never applied, because applying them was measured to be harmful or because the grammatical source provides no attested example. The lexicalized diminutive وکی, the last column of Table III, is one: it survives only in fixed words such as وړوکی, so removing it by rule damages them. Recording the affix and refusing to strip it is not the same as omitting it, because the inventory is also a description of the language. The minimum stem length does equally real work: without it the verb کېدل reduces to the single letter ک.

TABLE III. THE RULE SCHEMA, WITH THREE ENTRIES WRITTEN OUT

Field	ونه	توب	وکی
side	suffix	suffix	suffix
category	inflectional	derivational	derivational
min. stem	2	3	3
POS signalled	N	N	N
productivity	productive	productive	lexicalized
confidence	high	high	low
strip allowed	yes	yes	no
source	[T&R] M2	[N&K]	inventory §4
example	کورونه → کور	مینتوب → مین	وروکی unchanged

The third entry is documented and never applied. The header row and the last two rows are in Pashto script.

Table IV groups the 118 rules by what they do. Derivation accounts for 82 of them against 36 inflectional rules, which is the opposite of what a stemmer for English would need and follows from where Pashto puts its productive morphology. The grouping also shows where

the gated entries are concentrated: 17 of the 28 are noun-forming derivational suffixes, which is the area where lexicalization is commonest and where a rule is most likely to be documented in a grammar and still be unsafe to apply.

TABLE IV. THE 118 RULES BY FUNCTION

Group	Rules	Published	Kept	Examples
Nominal inflection	23	23	0	ونه انوې
Verbal inflection	13	11	2	بدل ول
Derivation: nouns	57	55	17	توب تیا تون

Group	Rules	Published	Kept	Examples
Derivation: adjectives	14	12	4	يز مند ناک
Derivation: verbs	1	1	1	—
Prefixes	10	10	4	هم بيا نيم
Total	118	112	28	

“Published” counts the rules citing a published grammar or lexical resource; “kept” counts those documented and never stripped. The full inventory is Appendix A.

C. Length rules

Three rules were contributed by the author, a native speaker of Pashto, from his own judgement, and then tested against the annotated data. A word of three letters or fewer is returned unchanged. A word of four or five letters ending in one of the five ye letters or in δ loses that letter.

A word ending in δ ستان loses that suffix, and stripping stops there.

The last condition is necessary: allowing the general rules to continue produced ترک → ترکمنستان rather than ترکمن. The second rule originally included ل, which we removed after measuring it. On a verb ل is the infinitive ending, but on a noun it is part of the root, and the rule could not distinguish the two: it reduced کابل to کاب and لام to لامل, accounting for 51 over-strips on its own.

A fourth rule was proposed and rejected. It would have removed the last three characters of any non-Pashto word of seven characters or more. Measured against three independent references, removing a documented affix instead was more accurate in both careful annotations (45.0% against 25.0% on the development set, 51.6% against 43.5% on the held-out set). Every word the rule damaged carried a genuine affix: مسوولیت became مسوو where the affix is یت, and اعلانات became اعلا where the affix is ات. The rule fired before the inventory was consulted and cut through the root.

D. Candidate generation and selection

Generation and selection are separate, which is the central design decision borrowed from the Persian and Urdu hybrid systems. The generator applies every rule that fits and returns all distinct forms reached, including the original word with nothing removed. For کورونو the pool contains the untouched form, the result of removing ونو, and the result of removing و alone. Nothing is decided at this stage, which is what allows a shallower analysis to displace a deeper one.

A word that is already a stem is not passed through untouched by default. Below the length rules it enters the same competition as any other word, as the no-strip candidate, and it can lose: that is why the system modifies 68.9% of held-out types while the reference modifies 62.7%, and why the do-nothing floor is not a floor the system inherits.

Selection scores each candidate on how much affix material was removed, whether other words in the corpus are built on the same stem, and whether the stem itself is attested. For a candidate reached by removing k characters the score is βk , plus a bonus F when the remainder has a word family of at least two members, plus either the attestation bonus A and w times the candidate's normalised corpus frequency if the stem is attested, or γk if it is not. The family term fires only when the remainder is at least three characters long.

TABLE V. WEIGHTS IN THE SELECTION SCORE

Symbol	Meaning	Value
β	reward per character removed	0.50
F	bonus when the remainder has a word family	2.00
$A (k>0)$	attestation bonus for a stripped candidate	2.00
$A (k=0)$	attestation bonus for the unstripped word	0.50
w	weight on corpus frequency	0.50
γ	reward per character for an unattested strip	0.20

Set by hand and adjusted on the development set; none were fitted automatically, and the held-out set was not used to choose any of them.

Two of these carry the design decisions. The attestation bonus for the unstripped word is a quarter of the one a stripped candidate gets, because the unstripped word is nearly always in the corpus and giving it the full bonus made inaction the default winner. And γ is above zero so the rules can act on a word the lexicon has never seen: an unattested strip still scores above nothing, so out-of-vocabulary input is stemmed rather than returned whole, though it loses to any attested candidate.

Two corrections to this stage were necessary during development and are worth recording, since both are easy to get wrong.

First, an early version required the stem to be an attested Pashto word before accepting a strip. That is incorrect: the output of a stemmer is a class label, and Porter's *relat* is not an English word either. Attestation now raises a candidate's score without gating it, and the system freely produces stems that are not words, including پوهن and لوبغاړ, ښوون.

Second, the unstripped input word is almost always present in the corpus, so awarding it the full attestation bonus made inaction the default winner and concealed every single-letter inflection. Presence in a corpus is evidence that a string is a word, not that it is a stem, and the no-strip candidate now receives a much smaller bonus.

A small number of affix groups bypass the scorer entirely; we call these the uniform-strip group, and it is the component measured under that name in Table X. The abstract-noun and place suffixes تون and والى, توب, تيا are long and unambiguous, so no competing analysis exists for the scorer to weigh; leaving them arbitrated meant that پوهنتون and روغتيا remained unchanged because روغ and پوهن are absent from the lexicon.

E. Throughput

The system is a single pass of string operations over a 118-rule table, with no model to load. On one core of a commodity laptop it stems roughly 6,800 word types per second cold and 7,100 warm, measured over the development set. For an indexing pipeline the stemmer is not the bottleneck.

F. Guards

Two guards were added after error analysis. A prefix is removed only when what remains is itself an attested word. Without it the system strips لا from لانسيت, leaving نسيت, and from لانتها, leaving انتها; with it, both words

are returned whole. The guard is conservative in both directions, since لانتها + لا genuinely is لانتها, and we accept the missed strip in exchange for not cutting into loanwords.

The second guard concerns the interaction between the prefix rules and the length rules. A word of four or five letters that begins with a removable prefix is passed to the full inventory instead of simply losing its last letter. Only a prefix that may actually be removed counts: نا is kept rather than stripped, so ناوړه is still handled by the length rule and becomes ناوړ, which is what the annotation gives.

A third component was implemented and then removed. A compound splitter returned the head of a word it judged to be a compound, which conflicted with the annotation policy's decision to keep compounds whole. Measured, it fired on seven types across the two evaluation sets and was wrong on all seven, usually because the "head" it returned was itself a suffix: نړىوال.

→ پتروليم and وال → يم. Disabling it is worth 0.22 points on the development set and 0.20 on held-out text. Splitting Pashto compounds reliably needs a lexicon of free forms that we do not have, and guessing at the boundary was worse than leaving it alone.

G. The verb dictionary and the scope of the task

The system includes a dictionary of 38 strong verbs covering 315 surface forms, of which 23 are checked against a printed grammar and 15 are marked unverified. The dictionary maps an inflected form to its verb: شو → کېدل. That output is usually the infinitive — 182 of the 315 forms come back as the infinitive and the remaining 133 as a stem, because the ل is then removed by a suffix rule — and an infinitive is a lemma, so substituting one word for another is lemmatization rather than stemming. On the word types whose reference is a truncation, the dictionary decides the output on 69 of them and is right on 11, because it supplies the wrong kind of answer; it costs 1.36 points on the development set and 0.80 on held-out text.

The dictionary is therefore disabled by default and provided as an explicit lemmatization mode. We report its contribution separately in Section VII rather than folding it into the stemming result.

5. DATA AND ANNOTATION

No annotated set for Pashto stemming was available, so two were constructed. The distinction between them determines how the results should be read.

A. Corpus

Word frequencies were taken from a corpus of Pashto news and general text assembled from VOA Pashto, BBC Pashto and a topic-labelled collection, totalling approximately 5.4 million tokens and 68,850 distinct word types. The lexicon shipped with the system is a 5,443-type frequency list drawn from this material.

Building a larger lexicon from the full corpus was attempted and made results worse. The reason is instructive: a raw corpus contains inflected forms as well as base forms, so with 48,000 types the attestation signal ceased to mean “this is a stem” and began rewarding over-stripped forms that happened to occur. The smaller list was better by being sparser.

B. Development set

2,912 word types were annotated independently by three native speakers of Pashto before the rule inventory was settled: the author, the principal of a Pashto-medium school, and a university teacher of the language. None of the three saw another's work, and none saw any system output. The author then adjudicated the disagreements and produced the released reference.

Two of the three passes are released as separate files, and the agreement between them can therefore be recomputed rather than asserted. On the 2,725 types both annotated, they chose the same stem for 96.88%, which is a Cohen's κ [14] of 0.969. Reduced to the decision that matters most — whether the word carries an affix at all — they agree on 98.24%, $\kappa = 0.965$. The two coefficients answer different questions. Chance agreement on the full stem string is 0.001, because the label space is every substring of the word, so there κ adds little to the raw figure; on the binary decision chance agreement is 0.503, and the κ is doing real work. Both fall in the band Landis and Koch [15] describe as almost perfect agreement.

The disagreements are small in substance as well as in number. Of the 86 types the two annotators stem differently, 84% differ by one or two characters: *مرست* against *مرس خون* against *خو*. They are disagreements about how deep to cut, not about which analysis applies.

The adjudicated reference stays close to both passes: 93.64% identical to the school principal's and 94.98% to the university teacher's. Where it departs from both, it

follows the written annotation policy, and the 42 policy decisions taken during the project are logged individually with their reasons.

One part of the construction is circular and we state it rather than leave it to be found. 116 verb forms were revised for consistency with the verb analysis, and 81 of those were changed to the reading the system itself produces, which is 3.0% of the stemming types. That is a real dependency between the reference and the system, it is confined to verbs, and it is why the held-out set rather than this one is treated as the result.

C. Held-out set

500 word types were sampled at random from the news corpus under three constraints: a frequency of at least five, to exclude typographical noise; a length of at least three characters; and no overlap with the development set. The sample was annotated by the same three native speakers, with the reference field initially empty so that nothing anchored them to any system's output. All three agreed on the result, and it is released as a single reference column. The set was not consulted during development, and every figure reported for it comes from a single run of the finished system.

The first pass was audited by comparing every removal against the affix inventory. On the released first-pass file, 73 of 327 removals, 22.3%, match no documented affix: words such as *مزاج* and *حارث*, which carry no affix at all, had been shortened. Thirty rows were revised in response, and the figure falls to 54 of 313, or 17.3%. The same audit on the development set leaves 108 of 1,370 removals, or 7.9%, unaccounted for.

That audit procedure is reusable and we recommend it. For each annotated pair, the material removed is compared against the affix inventory; a removal matching no affix is either an annotation error or evidence that the inventory is incomplete, and inspection separates the two. Applied to the development set, the same procedure identified six affixes the inventory lacked.

We went through all 54 removals that the audit leaves unaccounted for in the final held-out set, and the split is informative. Twenty-nine of them take material off one end of the word. Nineteen of those are gaps in the inventory rather than errors in the annotation, and most are variants of affixes already held: the oblique infinitive *ولو* beside *ول*, the oblique *ګرو* beside *ګر*, the feminine *يزې* beside *يزه*, the past participle *بدلې* beside *بدل*. Two more are combinations the engine cannot form because it strips

one layer at a time: ونو + توب is توبونو, and both halves are in the inventory. Eight are genuine annotation errors, and they cluster where one would expect — the English loans ايمرجينگ and ريسايکلنگ, where نگ is not a Pashto affix, the personal name ضيالرحمان, the place name پچيراگام. Two are ambiguous.

The remaining twenty-five take material off both ends at once, and all of them are verbs: ونبيي → ونبسي رارسېدو, نبيي → رسېد. The ending is in the inventory in every case; the prefix is a perfective or directional particle, and those are not held, for the reason Section IV.A gives — measured against both sets they were right between 13% and 50% of the time. These rows are therefore neither annotation errors nor an oversight in the inventory but the cost of that decision, stated here rather than left out of the count. An earlier version of the audit script reported only one-ended removals and so did not list them at all.

That the one-ended errors run three to one towards missing affixes rather than bad annotation is the useful part: it says the reference is sounder than the rule list, and it names the nineteen entries that would close the gap. The classified list is released with a verdict and a reason on every row.

D. Annotation policy

The policy the annotations follow is released with the data. Its main provisions are that the output is a stem and need not be a word; that Arabic broken plurals are handled by dictionary lookup rather than by rule; that proper nouns receive no special treatment, so ستان افغانستان loses its ستان like any other word; that compounds are kept whole; and that where two analyses are both defensible, the shallower is preferred.

6. EVALUATION PROTOCOL

A. Why exact-match accuracy alone is insufficient

The natural measure is the proportion of word types for which the system produces the annotated stem. On its own it is misleading. Nearly half of Pashto word types carry no affix and should be returned unchanged, so a system that modifies nothing scores 47.77% on the development set and 37.27% on the held-out set. Every letter a system declines to remove earns it credit.

Tuning against overall accuracy therefore drives a stemmer towards inactivity. During this work an apparent improvement of nearly three points proved, on inspection, to be exactly that: the system had become more conservative, losing seven points on the word types requiring a strip while gaining twelve on those requiring none.

All results below report three figures: accuracy on the types the annotation strips, accuracy on the types it leaves unchanged, and overall accuracy. The first cannot be obtained by inaction.

B. The scope of the stemming task

Stemming removes characters; it cannot insert them. Where the reference is an Arabic broken plural or a suppletive form — نجلي → نجونو, اثر → اثارو — the answer is a lemma that no affix rule can produce. Marking a truncating stemmer wrong on those types penalises it for not being a lemmatizer.

We therefore score the two tasks separately. A type belongs to the stemming task when the reference can be obtained by removing characters from the word, comparison being insensitive to which letter is written. By this criterion the development set contributes 2,717 stemming types and 195 lemma types, and the held-out set 499 and 1. Of the 195, 155 are verb forms whose reference is an infinitive that no truncation reaches; the remaining 40 are suppletive or irregular nominals and Arabic broken plurals, so the boundary between the two tasks is almost entirely the strong verb. Results for the lemma types are reported separately in Section VII.

C. Two protocols compared

Aslamzai and Saad report 87% accuracy; our reimplementations of the same nine rules reaches 40.48% on held-out data. The difference is entirely methodological, and the point is important enough to measure rather than assert.

Protocol A, used throughout this paper, fixes the reference before any system output is seen and counts a result only if it matches exactly. Protocol B reproduces theirs: a judge is shown the output and decides whether it is an acceptable analysis. We implement Protocol B reproducibly by counting an output as acceptable when nothing was removed that is not a documented affix, and by treating an unchanged word as always acceptable, which is how a judge shown an unchanged word would score it.

TABLE VI. THE SAME 499 HELD-OUT WORD TYPES UNDER BOTH PROTOCOLS

System	A: exact	B: judged	Modified
Modifies nothing	37.27%	100.00%	0.0%
Aslamzai & Saad [1]	40.48%	80.76%	37.5%
Proposed	72.75%	98.60%	68.9%

The final column gives the proportion of types the system modifies at all.

The first row of Table VI is the finding. Under Protocol B a system that modifies no word scores 100%, because every unchanged word is acceptable to a judge and it produces nothing else. The protocol cannot separate a working stemmer from an inert one, and a figure obtained under it cannot be compared with an exact-match figure.

Nothing here shows the published 87% to be incorrect. It shows that the measurement answers a different question. We note that the proposed system also

outperforms the baseline under Protocol B, by 98.60% to 80.76%, so the conclusion does not depend on the choice.

7. RESULTS

A. Held-out evaluation

Table VII reports the held-out set, which is the primary result: 499 word types from news text, annotated with the reference field initially empty and never used during development.

TABLE VII. HELD-OUT EVALUATION, 499 WORD TYPES

System	Accuracy	Affixed	Bare	UI
Modifies nothing	37.27%	0.0%	100.0%	1.000
Aslamzai & Saad	40.48%	16.9%	80.1%	0.889
Proposed	72.75%	72.8%	72.6%	0.500

“Affixed” and “Bare” are accuracy on the types the annotation changes and leaves unchanged.

The two systems fail in opposite directions, and that is more informative than the gap between them. Of the baseline’s 297 held-out errors, 254 — 86% — are under-strips, and 163 are words it left untouched altogether; only 9% are over-strips. Of our 143, 51% are over-strips and 44% under-strips. The baseline’s problem is inaction, ours is excess, which is the trade a stemmer has to make and the reason the two halves of the task are reported separately.

The proposed system improves on the baseline by 32.3 points overall. The separation of the two halves is more informative than the overall figure: on the types that require a strip the baseline reaches 16.9% and the proposed system 72.8%, a factor of four. Paice’s under-stemming index falls from 0.889 to 0.500, and over-stemming remains below 4×10^{-5} .

B. Development set

TABLE VIII. DEVELOPMENT SET, 2,717 STEMMING TYPES

System	Accuracy	Affixed	Bare	UI
Modifies nothing	47.77%	0.0%	100.0%	1.000
Aslamzai & Saad	50.68%	12.2%	92.8%	0.919
Proposed	81.19%	81.0%	81.4%	0.291

Table VIII reports the development set. The figure is higher than the held-out result. The revisions described in Section V.B account for 0.07 points of that gap, so the explanation lies elsewhere: it is orthographic, and Section VII.E gives it. The development set is reported for comparability with the baseline, which is scored identically, and because Paice’s indices are more stable on the larger set. On the types requiring a strip the margin is wider still: 81.0% against 12.2%, a factor of six.

C. Two external datasets

After the system was finalised, it was run against two independently annotated Pashto word lists of 5,000 and 10,000 types. Neither was produced by this project, and 96% of their entries appear in neither of our evaluation sets. Exact-match accuracy is 72.30% and 70.18%, against 51.24% and 47.95% for the baseline. Those figures sit beside the 72.75% measured on held-out news text, which is the point: the system does not depend on the data it was built against.

TABLE IX. TWO INDEPENDENTLY ANNOTATED WORD LISTS

Dataset	Types	Exact	Folded
External, 5,000 types	5,000	72.30%	80.14%
Aslamzai & Saad		51.24%	57.36%
External, 10,000 types	10,000	70.18%	77.17%
Aslamzai & Saad		47.95%	53.68%
Our held-out set	499	72.75%	73.55%
Our development set	2,717	81.19%	82.63%

The folded column compares with the five ye letters treated as equal; the stemmer's output is never rewritten.

Both lists carry a reference column normalised to ى while their word column retains ي, so a stem is scored wrong whenever it preserves the letter the word actually has. Folding the five ye letters together for the comparison alone raises the two figures to 80.14% and 77.17%. The same folded comparison is worth 1.44 points on our development set and 0.80 on our held-out set, both of which are internally consistent in their orthography. The size of that difference is what the measurement shows: seven to eight points on the external lists against one on ours places the disagreement in how they spell the reference rather than in the stemming.

One caveat limits what these lists establish. On the words they share with our development set the two references agree 70.8% and 64.1% of the time. Sorting the disagreements shows they are a difference of policy rather

than noise. About a quarter — 19.1% on the 5,000-type list and 24.3% on the larger one — are cases where our reference gives a lemma that no truncation reaches, اخی

→ اخيستل against their اخ; those are not disagreements about stemming at all but about where stemming stops. Of the rest, roughly half cut deeper than we do (تنکى → تن against our تنگ) and about a quarter cut less (استازيتوب)

→ استازى against our استاز). Only two disagreements in 154 come down to the spelling of a ye. The two annotations differ consistently in how deep a stem should go and in whether verbs are reduced to a stem or to an infinitive, which is why these lists test generalisation rather than correctness under our policy.

D. Component study

TABLE X. EACH COMPONENT REMOVED IN TURN

Configuration	Development	Held-out
Full system	81.19%	72.75%
– suffix rules	71.95% (–9.24)	59.12% (–13.63)
– length rules	79.39% (–1.80)	70.54% (–2.20)
– corpus lexicon	80.49% (–0.70)	72.75% (+0.00)
– uniform-strip group	80.82% (–0.37)	72.55% (–0.20)
– prefix rules	81.16% (–0.04)	72.95% (+0.20)
+ verb dictionary	79.83% (–1.36)	71.74% (–1.00)

Changes in percentage points.

The suffix inventory does most of the work, and removing it costs 12.4 points on held-out text. The author's length rules contribute between two and three points on both sets.

Three rows record decisions taken against the measured score, and we state them explicitly rather than quietly adopting whichever setting scored best.

Prefix stripping costs a few hundredths of a point on the development set. We retain it because هم, بيا, نيم and لا are genuine Pashto prefixes, and removing the rules would amount to claiming the language has none; the attestation guard described in Section IV.E is what makes them safe. The small effect is itself explained by the inventory: of the ten prefix rules, four are documented and

never stripped. نا, غير and بې are negation prefixes that the annotation policy keeps, because ناقانونه is not an instance of قانون but its opposite, and سر is a free noun as often as it is a prefix.

The corpus lexicon is worth 0.18 points on the development set and –0.40 on held-out text. Its contribution is now marginal, because the uniform-strip and length rules decide most cases before the scorer is consulted. We report this plainly: the system should not be described as lexicon-driven.

Enabling the verb dictionary costs 1.36 points on the development set and 0.80 on held-out text, for the reason given in Section IV.F. Its proper contribution is to the

lemma types, where it answers 51 of the 195 in the development set against 8 without it.

E. Orthography

The gap between the two evaluation sets has one dominant cause. Of the 500 held-out news types, 118 contain the hard ye ې; of the 2,912 development types, none do. The inventory contained thirteen rules ending in the narrow ye ى and none ending in ې, so a large class of news words passed through untouched: لومړي and تاريخي were returned unchanged.

Making rule matching insensitive to which ye is written, while leaving the output letters untouched, corrected this. On the development set, which is written entirely in one convention, exact matching scores 0.15 points higher; on held-out news text the folded comparison is worth 1.20 points, so it earns its place on

the set that reflects real text. This is the clearest instance in the project of a defect that a single-source evaluation set could not have revealed.

8. DISCUSSION

A. Correct rules and under-determined rules

A per-rule analysis separates two kinds of failure, and the distinction matters for how the remaining error should be addressed.

Table XI gives the precision of the rules that fire most often on the development set: how many times each contributed to the analysis finally selected, and how often that analysis matched the reference. Publishing it is uncomfortable and necessary. A rule can contribute positive points overall and still be wrong half the time it fires, and a reader cannot tell the two apart from an ablation table.

TABLE XI. PRECISION OF THE FOURTEEN MOST FREQUENTLY FIRING RULES

Rule	Fires	Correct	Precision
R2 keep short word	398	357	89.7%
R1 drop ه	227	161	70.9%
R1 drop ى	190	149	78.4%
R1 drop ې	160	133	83.1%
– ى	144	129	89.6%
– و	132	109	82.6%
final trim	92	50	54.3%
– ونو	58	54	93.1%
– ه	47	30	63.8%
– ونه	40	36	90.0%
R2 drop ې at 3 letters	39	21	53.8%
R1 drop ى	36	31	86.1%
– ان	36	15	41.7%
R2 drop و at 3 letters	32	18	56.2%

“Fires” counts the analyses finally selected in which the rule took part. Development set. Generated by experiments/error_analysis.py.

Building that table changed the system. Two rules turned out to fire 31 times between them and be right on none of those occasions: لو, the oblique infinitive, and لې, the feminine past participle. Both bundle an ل into the affix that the reference keeps, either as the infinitive marker (ساتلو → ساتل, not سات) or as part of the root (بريالی → بريال). The rules that remove the final vowel alone already produce the right answer, so these two could only ever take one letter too many. They are now documented and never applied, which is worth 1.07 points on the development set and nothing either way on held-out text.

This is the same defect as the standalone verbal ل, removed earlier for the same reason: it applied only to nouns where ل belongs to the root and turned کابل into کاب. Two further rules were removed during the work, one taking ى as a unit where the affix is usually the ى alone, and one restoring a feminine ه to any stem that had lost one, which produced کورونه → کورونو.

Four rules remain below 60% precision and are kept. ان is right 15 times in 36, but it is the ordinary animate plural of Pashto and gating it costs 0.40 points on held-out

text; its low precision reflects competition with longer analyses rather than a defect in the rule. The two three-letter length rules sit at 53.8% and 56.2% for the reason the next paragraph describes, and the alternative to a coin-flip there is to leave every short inflected word unstemmed.

Other rules are correct but under-determined. The rule removing a final ښ from a short word fires 227 times on the development set and is correct 161 of those. لنډه → لنډ is right and خواره → خوار is wrong, and the two words are indistinguishable in shape. Removing the rule loses more than it saves: overall accuracy falls by 1.44 points. No reformulation conditioned on the surface form can help; only a lexical resource can.

B. What limits the current accuracy

The 511 development-set errors divide into three kinds. 305 of them, 60%, are over-strips, where the system removed more than the reference does; 167, 33%, are

under-strips, and 75 of those are words the system left untouched altogether; the remaining 41 are the same length as the reference but differ in which letters survive. The held-out set behaves the same way, 54% and 41%. Over-stripping is therefore the larger problem on both sets, which is what one would expect of a system whose scorer rewards removing affix material.

Table XII locates the errors. Each type is assigned to the group of the rule that explains the reference's removal, under the same grouping as Table II, so that the two tables can be read against each other. Accuracy is high wherever the reference removes an affix the inventory holds — 88.8% on nominal inflection, 83.5% on noun-forming derivation, 95.1% on verbal inflection — and the two weak rows are the ones that do not depend on the inventory being right. 108 types have a reference the inventory cannot explain at all, and we reach 29.6% on those. The larger group is the 1,347 types the reference leaves whole, where we are right 78.4% of the time: 291 words stripped that should not have been, the biggest single source of error in the system.

TABLE XII. ACCURACY BY THE AFFIX THE REFERENCE REMOVES

What the reference removes	Types	Accuracy
Nominal inflection	1,116	88.8%
Verbal inflection	41	95.1%
Derivation: nouns	79	83.5%
Derivation: adjectives	26	84.6%
No affix, word left whole	1,347	78.4%
Nothing in the inventory	108	29.6%

Development set. The grouping is the one in Table II. Generated by `experiments/error_analysis.py`.

We had attributed that bucket to loanwords, and the data does not support it. Pashto marks native vocabulary orthographically: a word containing one of ټ ډ ږ ښ ځ څ is certainly not a borrowing, though a word without one may be either, so the test is one-sided. Among the types the reference leaves whole we are wrong on 28.3% of the certainly-native ones and 18.8% of the rest. Native words are the harder half, not the easier one, and the errors bear it out: خواره, وړاندې, داسې, پاتې, يوازې are ordinary Pashto words ending in ښ or ې that the length rules cut. The shortfall is the ښ/ې/ي ambiguity, and it falls on native vocabulary rather than on borrowings.

We tested the obvious remedy and report the result because it is negative. A 257-entry list of development-set words that should be left unchanged raises development accuracy by 8.5 points and held-out accuracy by zero: of the 51 held-out types needing the same treatment, none appear in the list. The list is pure memorisation of one

sample. The generalizing form of the same idea is a lexicon of words that must not be stripped. On the evidence above it would need to cover ordinary native vocabulary rather than borrowings alone, which makes it a larger undertaking than we first assumed, and we leave it to future work.

9. LIMITATIONS

The evaluation is on word types rather than running text, so no measurement of retrieval effectiveness is reported, and the relationship between the intrinsic gains shown here and downstream performance remains untested.

Of 118 affix rules, 112 cite a published grammar by chapter or section; the six that do not are entries the engine never applies. The chapter attributions were assembled by the author and have not been independently checked page by page, so a reader verifying the inventory should treat each locus as a pointer rather than as a quotation. Of 38 verbs in the dictionary, 23 are checked against a printed grammar and 15 are marked unverified in the released

documentation. We prefer to label these rather than to present them as settled.

Compounds are kept whole, which is a policy choice rather than a solution: it avoids a class of error at the cost of never conflating لاسلیک with لاس. The annotation does not follow the policy perfectly either, and an audit during development found types where the reference itself splits a compound. The released audit script reports every removal that matches no documented affix, which is where such cases surface.

The corpus is drawn from news and general text, so the vocabulary of specialised domains is under-represented.

10. CONCLUSION

We presented a rule-based stemmer for Pashto built from an affix inventory of 118 rules taken from published grammars, each recording its source, its productivity and an explicit decision about whether it may be applied. No machine learning is used, and every decision the system makes can be traced to a rule or a dictionary entry.

On 499 held-out word types from news text, annotated blind and never used during development, the system reaches 72.75% exact-match accuracy against 40.48% for the published baseline, and 72.8% against 16.9% on the types that require a strip. On a 2,717-type development set it reaches 81.19% against 50.68%, with Paice's understemming index falling from 0.919 to 0.291.

Three observations generalise beyond this particular system. Exact-match accuracy is not a sufficient measure for stemming in a language where a large share of word types carry no affix, because it rewards inactivity; reporting the two halves of the task separately costs nothing and prevents the error. An evaluation set drawn from a single orthographic convention will conceal defects that a second source exposes immediately, as the ye variants did here. And when a rule-based system finds no affix, the useful response is to ask what the inventory is missing rather than to remove characters regardless: the one rule in this work that guessed a boundary instead of finding one was measured against three references, lost on both careful annotations, and was removed.

The stemmer, the affix inventory with its sources, the verb dictionary, both annotated sets and the scripts that reproduce every figure in this paper are released under an MIT licence at https://github.com/Khairullah-Ibrahim-Khail/Pashto_Stemmer.

Every table in this paper is regenerated by `experiments/reproduce_paper.py`, and the inter-annotator figures by `experiments/agreement.py`.

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REFERENCES

- [1] S. Aslamzai and S. Saad, "Pashto language stemming algorithm," *Asia-Pacific Journal of Information Technology and Multimedia*, vol. 4, no. 1, pp. 25–37, 2015.
- [2] M. F. Porter, "An algorithm for suffix stripping," *Program*, vol. 14, no. 3, pp. 130–137, 1980.
- [3] J. B. Lovins, "Development of a stemming algorithm," *Mechanical Translation and Computational Linguistics*, vol. 11, pp. 22–31, 1968.
- [4] C. D. Paice, "Another stemmer," *ACM SIGIR Forum*, vol. 24, no. 3, pp. 56–61, 1990.
- [5] C. D. Paice, "An evaluation method for stemming algorithms," in *Proc. 17th Annual International ACM SIGIR Conference*, 1994, pp. 42–50.
- [6] S. Khoja and R. Garside, "Stemming Arabic text," Computing Department, Lancaster University, Tech. Rep., 1999.
- [7] L. S. Larkey, L. Ballesteros, and M. E. Connell, "Improving stemming for Arabic information retrieval: light stemming and co-occurrence analysis," in *Proc. 25th Annual International ACM SIGIR Conference*, 2002, pp. 275–282.
- [8] A. A. Sharifloo and M. Shamsfard, "A bottom up approach to Persian stemming," in *Proc. 3rd International Joint Conference on Natural Language Processing*, 2008.
- [9] H. Tegey and B. Robson, *A Reference Grammar of Pashto*. Washington, DC: Center for Applied Linguistics, 1996.
- [10] B. Robson and H. Tegey, "Pashto," in *The Iranian Languages*, G. Windfuhr, Ed. London: Routledge, 2009.
- [11] A. B. David, *Descriptive Grammar of Pashto and its Dialects*. Berlin: De Gruyter Mouton, 2014.
- [12] H. Penzl, *A Grammar of Pashto: A Descriptive Study of the Dialect of Kandahar, Afghanistan*. Washington, DC: American Council of Learned Societies, 1955.
- [13] T. Naeem and M. A. Khan, "Automatic derivation of nouns from adjectives in Pashto," *Center for Language Engineering, Pakistan*.
- [14] J. Cohen, "A coefficient of agreement for nominal scales," *Educational and Psychological Measurement*, vol. 20, no. 1, pp. 37–46, 1960.
- [15] J. R. Landis and G. G. Koch, "The measurement of observer agreement for categorical data," *Biometrics*, vol. 33, no. 1, pp. 159–174, 1977.
- [16] A. Khan, "The diminutive morphological function between English and Pashto," *Humanities and Social Sciences Communications*, vol. 10, art. 536, 2023.

APPENDIX A: THE COMPLETE AFFIX INVENTORY

The 118 rules in full, in the order the engine holds them. "Type" gives the side and whether the rule is inflectional or derivational; "min" is the shortest stem the rule is allowed to leave; "strip" is whether the engine may remove the affix at

all; “source” is what backs the rule — a published grammar, this project's inventory document, and gated for an entry that is never applied and appeals to no grammar. The entries marked no under “strip” are documented and never applied, for the reasons given in Section IV.B. This table is generated directly from the rule objects by `paper/make_inventory_tables.py`, so it cannot drift from the implementation.

TABLE A.I. THE COMPLETE AFFIX INVENTORY

Affix	Type	min	POS	Productivity	Confidence	Strip	Source
<i>Nominal inflection</i>							
تانه	suf, infl	3	N	productive	high	yes	W
تانو	suf, infl	3	N	productive	high	yes	W
گانو	suf, infl	2	N	productive	high	yes	T&R M4/F3
گانې	suf, infl	2	N	productive	high	yes	T&R F2/F3
گونه	suf, infl	2	N	productive	high	yes	T&R ch.5 collective numbers
گونو	suf, infl	2	N	productive	high	yes	T&R ch.4 noun inflection
اتو	suf, infl	3	N	productive	high	yes	T&R ch.4 loanword case paradigm
انو	suf, infl	2	N	productive	high	yes	T&R M1
انې	suf, infl	3	N	productive	high	yes	T&R F2
جات	suf, infl	3	N	productive	high	yes	T&R ch.2 Persian nominal suffixes
ونه	suf, infl	2	N	productive	high	yes	T&R M2
ونو	suf, infl	2	N	productive	high	yes	T&R M2
گان	suf, infl	2	N	productive	high	yes	T&R ch.4 animate plurals
ات	suf, infl	3	N	productive	high	yes	T&R ch.2 Arabic borrowings
ان	suf, infl	3	N	productive	high	yes	T&R M1
ون	suf, infl	3	N	productive	high	yes	N&K
وو	suf, infl	3	N	productive	high	yes	T&R
وې	suf, infl	3	N	productive	high	yes	T&R
يو	suf, infl	3	N	productive	high	yes	T&R F2/F3
ه	suf, infl	3	N/ADJ	productive	high	yes	T&R ch.4 gender, ch.5 agreement
و	suf, infl	3	N/ADJ	productive	high	yes	T&R M3/Adj2
ۍ	suf, infl	3	N/ADJ	productive	high	yes	T&R ch.4 feminine noun classes
ې	suf, infl	2	N/ADJ	productive	high	yes	T&R F1/Adj
<i>Verbal inflection</i>							
ېدا	suf, infl	2	V	productive	high	yes	R&T 13.16
ېدل	suf, infl	2	V	productive	high	yes	T&R ch.6 derivative verbs
ېده	suf, infl	2	V	productive	high	yes	T&R ch.6 imperfective past stems
ېدو	suf, infl	2	V	productive	high	yes	R&T 13.16
ېږم	suf, infl	2	V	productive	high	yes	R&T 13.16

Affix	Type	min	POS	Productivity	Confidence	Strip	Source
بزو	suf, infl	2	V	productive	high	yes	R&T 13.16
بزی	suf, infl	2	V	productive	high	yes	R&T 13.16
لو	suf, infl	2	V	productive	high	no	gated
ول	suf, infl	2	V	productive	high	yes	T&R ch.6 transitive derived verbs
بد	suf, infl	2	V	productive	high	yes	R&T 13.16
بزی	suf, infl	2	V	productive	high	yes	R&T 13.16
ئ	suf, infl	3	V	productive	high	yes	T&R ch.6 imperative paradigm
ل	suf, infl	2	V	productive	low	no	gated
Derivation: nouns							
یدونکی	suf, der	3	N/ADJ	productive	high	yes	R&T 13.16
بدونکی	suf, der	3	N/ADJ	productive	high	yes	R&T 13.16
خانه	suf, der	3	N	productive	high	yes	T&R ch.4 derived nouns of place
دانی	suf, der	3	N	borrowed	medium	yes	Penzl §51 containers
ستان	suf, der	3	N	borrowed	medium	no	inv §2.3
غالی	suf, der	3	N	lexicalized	low	no	T&R ch.4 nouns of place
والو	suf, der	3	N	productive	high	yes	T&R ch.4 oblique plural formations
والی	suf, der	3	N	productive	high	yes	N&K
ونکو	suf, der	3	N	productive	high	yes	T&R ch.6 participle case shifts
ونکی	suf, der	3	N/ADJ	productive	high	yes	T&R
ونکې	suf, der	3	N/ADJ	productive	high	yes	T&R ch.6 active participle paradigm
کوئی	suf, der	3	N	restricted	low	no	Penzl §53 diminutives
کلوی	suf, der	3	N	restricted	high	yes	T&R ch.4 abstract noun formations
گوئی	suf, der	3	N	restricted	low	no	Penzl §53 diminutives
اوی	suf, der	3	N	restricted	medium	no	T&R ch.6 verbal nouns
بان	suf, der	3	N	borrowed	high	yes	Penzl §52 suffixes of the doer
توب	suf, der	3	N	productive	high	yes	N&K
تون	suf, der	3	N	productive	high	yes	N&K
تیا	suf, der	2	N	productive	high	yes	T&R ch.4 noun derivation
خور	suf, der	3	N	borrowed	medium	yes	T&R ch.4 synthetic compounds
دار	suf, der	3	N	productive	high	yes	T&R ch.2 Persianate compounds
دان	suf, der	4	N	borrowed	medium	yes	Penzl §51 containers
زار	suf, der	3	N	borrowed	medium	yes	T&R ch.4 locational suffixes

Affix	Type	min	POS	Productivity	Confidence	Strip	Source
لیک	suf, der	3	N	productive	medium	no	T&R ch.4 administrative compounds
نکو	suf, der	3	N/ADJ	productive	high	yes	T&R ch.6 participle variations
نکی	suf, der	3	N/ADJ	productive	high	yes	T&R ch.6 active participle paradigm
نکې	suf, der	3	N/ADJ	productive	high	yes	T&R ch.6 active participle paradigm
وال	suf, der	3	N	restricted	medium	no	T&R ch.4 suffixes of affiliation
وان	suf, der	3	N	restricted	low	no	Penzl §52 suffixes of the doer
ولي	suf, der	3	N	restricted	low	no	T&R ch.4 native abstract derivations
وتی	suf, der	3	N	productive	high	yes	Khan23
وکی	suf, der	3	N	lexicalized	low	no	Penzl §53 diminutives
پال	suf, der	3	N/ADJ	productive	high	yes	T&R ch.4 synthetic compound stems
پوه	suf, der	3	N	productive	high	yes	T&R ch.4 compound nouns
ځای	suf, der	3	N	productive	high	yes	T&R ch.4 derived nouns of place
ځیو	suf, der	3	N	productive	high	yes	W
ښنه	suf, der	3	N	restricted	high	yes	T&R ch.4 action nouns from verbs
کار	suf, der	3	N	productive	high	yes	Penzl §52 suffixes of the doer
گار	suf, der	3	N	borrowed	high	yes	Penzl §52 suffixes of the doer
گره	suf, der	3	N	productive	high	yes	N&K
گری	suf, der	3	N	productive	high	yes	Penzl §52 suffixes of the doer
یال	suf, der	3	N	restricted	medium	no	T&R ch.5 suffixal adjectives
ار	suf, der	4	N	restricted	medium	no	T&R ch.4 noun derivations
اک	suf, der	4	N	restricted	high	yes	Penzl §53 diminutives
نه	suf, der	3	N	productive	high	yes	W
نې	suf, der	3	N	productive	medium	no	gated
ځی	suf, der	3	N	productive	high	yes	W
چي	suf, der	3	N	lexicalized	low	no	Penzl §52 suffixes of the doer
ښت	suf, der	3	N	productive	high	yes	W
گر	suf, der	3	N	productive	high	yes	Penzl §52 suffixes of the doer
کی	suf, der	3	N	productive	high	yes	Khan23
یت	suf, der	3	N	productive	high	yes	T&R ch.2 morphological loan layers
یي	suf, der	3	ADJ/N	productive	high	yes	T&R ch.5 noun-to-adjective suffixes
ا	suf, der	4	N	restricted	low	no	T&R ch.4 abstract suffixes
ت	suf, der	4	N	restricted	medium	no	T&R ch.2 loan morphology

Affix	Type	min	POS	Productivity	Confidence	Strip	Source
ک	suf, der	4	N	restricted	medium	no	T&R ch.3 consonant alternations
ی	suf, der	3	N/ADJ	productive	high	yes	W
Derivation: adjectives							
انه	suf, der	3	ADJ	productive	high	yes	T&R ch.5 suffixal derivations
مند	suf, der	3	ADJ	productive	high	yes	T&R ch.5 suffixal adjectives
ناک	suf, der	3	ADJ	productive	high	yes	T&R ch.5 adjective derivation
چکی	suf, der	3	ADJ	productive	high	yes	Khan23
یزه	suf, der	3	ADJ	productive	high	yes	T&R ch.5 modern formations
یزو	suf, der	3	ADJ	productive	high	yes	T&R ch.5 oblique adjectival agreement
جن	suf, der	3	ADJ	restricted	high	yes	T&R ch.5 noun-to- adjective suffixes
من	suf, der	3	ADJ	productive	high	yes	T&R ch.5 regional suffixal variants
نی	suf, der	3	ADJ	productive	medium	no	gated
نی	suf, der	3	ADJ	productive	medium	no	gated
ور	suf, der	3	ADJ	restricted	medium	no	T&R ch.5 suffixal adjectives
وړ	suf, der	3	ADJ	productive	high	yes	W
ین	suf, der	3	ADJ	restricted	medium	no	T&R ch.5 suffixal adjectives
یز	suf, der	3	ADJ	productive	high	yes	T&R ch.5 modern formations
Derivation: verbs							
لې	suf, der	3	ADJ/V	productive	high	no	T&R
Prefixes							
بلا	pre, der	3	ADJ	borrowed	medium	yes	T&R ch.5 intensive modifiers
بیا	pre, der	3	V/N	productive	high	yes	T&R ch.7 adverbs
غیر	pre, der	3	ADJ	borrowed	high	no	T&R ch.2 morphological borrowings
نیم	pre, der	3	ADJ/N	restricted	high	yes	T&R ch.5 numbers and modifiers
بې	pre, der	3	ADJ/N	productive	high	no	T&R ch.5 prefixed adjectives
سر	pre, der	4	N	restricted	medium	no	T&R ch.4 nominal compounding
ضد	pre, der	3	N	borrowed	medium	yes	T&R ch.2 modern loan vocabulary
لا	pre, der	3	—	borrowed	medium	yes	T&R ch.2 Arabic/Persian bound morphemes
نا	pre, der	3	ADJ/N	productive	high	no	T&R ch.5 prefixed adjectives
هم	pre, der	3	—	productive	high	yes	T&R ch.2 Persian compounds

APPENDIX B: THE IRREGULAR-VERB DICTIONARY

The 38 strong verbs the dictionary holds, with the 315 surface forms they map. These are the verbs whose present stem cannot be derived from the infinitive. 15 entries are marked unverified and are labelled as such here rather than presented as settled. The dictionary returns a lemma, so it is off by default and belongs to the lemmatization mode described in Section IV.G. This table is generated from the dictionary by `paper/make_verb_appendix.py`, so it cannot claim a verb the software does not hold.

TABLE B.I. THE IRREGULAR-VERB DICTIONARY

Infinitive	Present stem	Past	Forms	Gloss, source
کول	کو	کر / کاوه	23	to do/make — Robson & Tegey
کېدل	کېږ / کيږ	شو / شول	18	to become/happen — Robson & Tegey
تلل	ځ	لار / تل	16	to go — Robson & Tegey
راتلل	راځ	راغل / راغی	12	to come — Robson & Tegey
وړل	وړ	يووړ / يوړ	11	to carry/take away — Robson & Tegey
راوړل	راوړ	راووړ	6	to bring — Robson & Tegey
خوړل	خور	خوړ	9	to eat — Tegey & Robson (1996)
څښل	—	—	6	to drink — unverified
لیدل	وین	لید	10	to see — Tegey & Robson (1996)
کتل	گور	کوت / کت	8	to look — Tegey & Robson (1996)
اخیستل	اخل	اخیست	9	to take/buy — Tegey & Robson (1996)
نیول	نیس	نیو	8	to catch/hold — Tegey & Robson (1996)
ویل	وای	ویل	10	to say — Tegey & Robson (1996)
ورکول	ورکو	ورکر	9	to give — Robson & Tegey
غوښتل	غوار	غوښت	10	to want — Tegey & Robson (1996)
موندل	موم	موند	8	to find — Tegey & Robson (1996)
لوستل	لول	لوست	6	to read — Tegey & Robson (1996)
لړل	لړ	درلود / لړل	11	to have — unverified
وژل	وژن	وژل / ووژه	7	to kill — Tegey & Robson (1996)
اورېدل	اور	اورېد	8	to hear — unverified
پېژندل	پېژن / پیژن	پېژند / پیژند	7	to know/recognize — Tegey & Robson (1996)
رسېدل	—	—	9	to arrive / reach — unverified
پوهېدل	—	—	7	to understand — unverified
اوسېدل	—	—	6	to live / reside — unverified
ښودل	ښی	ښود	7	to show — Tegey & Robson (1996)
کېښودل	ږد	کېښود / ایښود	7	to put/place — unverified
پرېښودل	پرېږد	پرېښود	5	to leave/abandon — unverified
الوتل	الوز	الوت	6	to fly — Tegey & Robson (1996)

Infinitive	Present stem	Past	Forms	Gloss, source
ختل	خېژ	خوت / خت	6	to climb/rise — Tegey & Robson (1996)
پرېوتل	پرېوخ	پرېوت	5	to fall — unverified
کښېنستل	کښېن	کښېناست / ناست	6	to sit down — unverified
پاخېدل	—	—	5	to stand up / rise — unverified
گڼل	گڼ	گانه / گڼل	6	to count/consider — unverified
بلل	بول	بلل	6	to call/consider — Tegey & Robson (1996)
سپارل	—	—	6	to hand over / entrust — unverified
ژغورل	—	—	5	to save / rescue — unverified
ايستل	باس	ايست / وېست	6	to pull out/extract — Tegey & Robson (1996)
کارول	—	—	5	to use / operate — unverified

A further 12 strong verbs were identified during the work and are not in the dictionary. They are listed because a reader extending the dictionary should start here, and because an earlier version of this appendix presented them as though they were implemented.

TABLE B.II. IDENTIFIED BUT NOT IN THE DICTIONARY

Infinitive	Present stem	Past	Gloss	Source
اغوستل	اغوند	اغوست	to wear	Tegey & Robson (1996)
اوړل	اوړ	وېښت / اوښت	to pass/cross	Tegey & Robson (1996)
چاودل	چو	چاود	to explode	Tegey & Robson (1996)
نغښتل	نغاړ	نغښت	to wrap/roll up	Tegey & Robson (1996)
ويشتل	ول	وېشت	to shoot	Tegey & Robson (1996)
وتل	وخ	وت	to go out	unverified
رودل	رو	رود	to suck	Tegey & Robson (1996)
ورتل	ورخ	ورتل / ورغل	to go there	unverified
درتل	درخ	درتل / درغل	to go to you	unverified
راکول	راکو	راکر	to give here	unverified
درکول	درکو	درکر	to give to you	unverified
پرانیستل	پرانیز	پرانیست	to open	unverified